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Data Science For Life



Outline

- ❑ Independent Component Analysis
- ❑ The Cocktail Party (18+)
- ❑ Neural Cocktail Party



Independent Component Analysis

Linear Combination of Signals

Two observed signals $x_1(t)$ and $x_2(t)$, which are linear combinations of two sources $s_1(t)$ and $s_2(t)$:

$$x_1(t) = a_{11}s_1(t) + a_{12}s_2(t)$$

$$x_2(t) = a_{21}s_1(t) + a_{22}s_2(t)$$

where a_{ij} are the mixing coefficients.

Given a 2×2 matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

The formula for the inverse of a 2×2 matrix

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

provided that $ad - bc \neq 0$.



Independent Component Analysis

Model

$$x_1(t) = a_{11}s_1(t) + a_{12}s_2(t)$$

$$x_2(t) = a_{21}s_1(t) + a_{22}s_2(t)$$

We observe over time

$$\begin{pmatrix} x_1(1) & x_1(2) & \cdots & x_1(t) \\ x_2(1) & x_2(2) & \cdots & x_2(t) \end{pmatrix}$$

We want

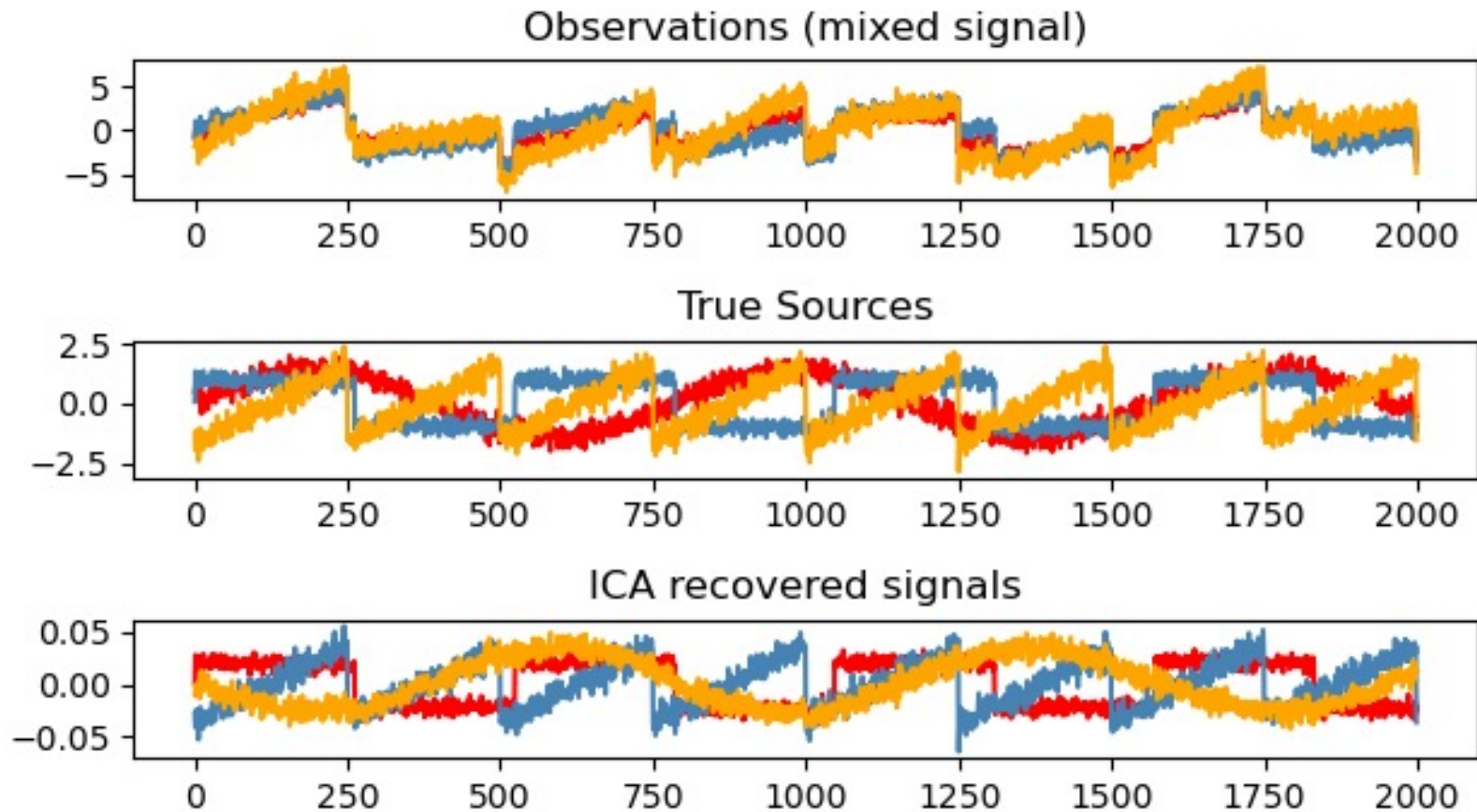
$$\begin{pmatrix} s_1(1) & s_1(2) & \cdots & s_1(t) \\ s_2(1) & s_2(2) & \cdots & s_2(t) \end{pmatrix}$$

Goal

Estimate $\{s_i(t)\}$, (and also $\{a_{ij}\}$).



Independent Component Analysis



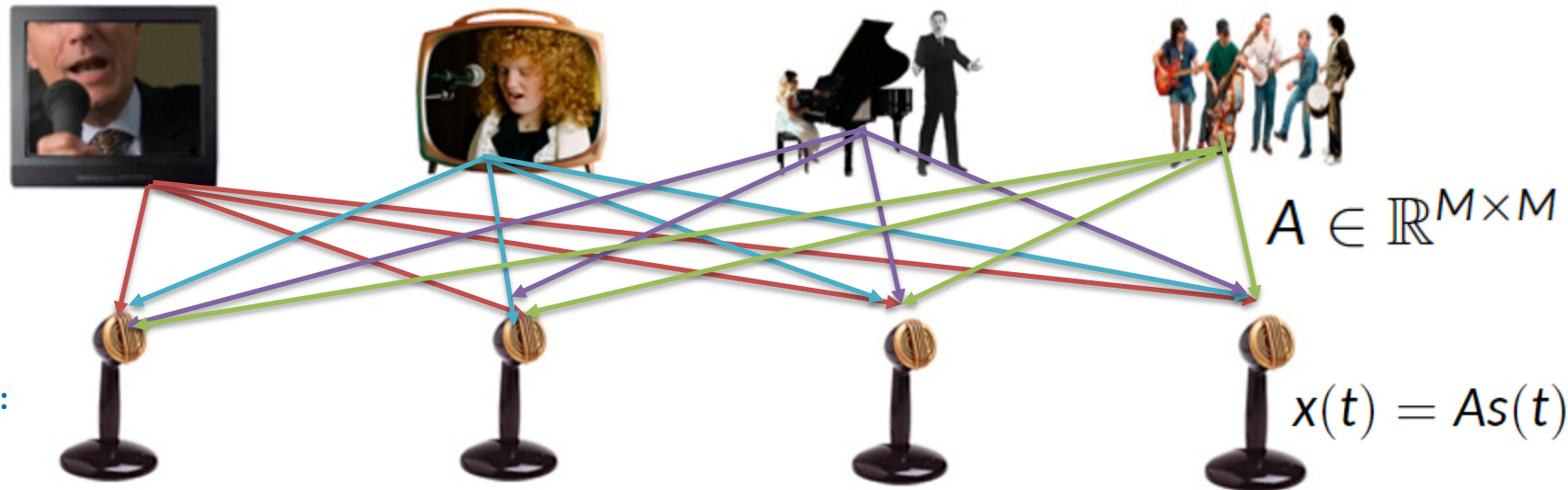
The Cocktail Party (18+)



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The Cocktail Party (18+)

Original Sources: $s(t)$



ICA Estimation: $y(t) = Wx(t)$



Some ICA Applications

STATIC

- Image denoising
- Microarray data processing
- Decomposing the spectra of galaxies
- Face recognition
- Facial expression recognition
- Feature extraction
- Clustering
- Classification
- Deep Neural Networks

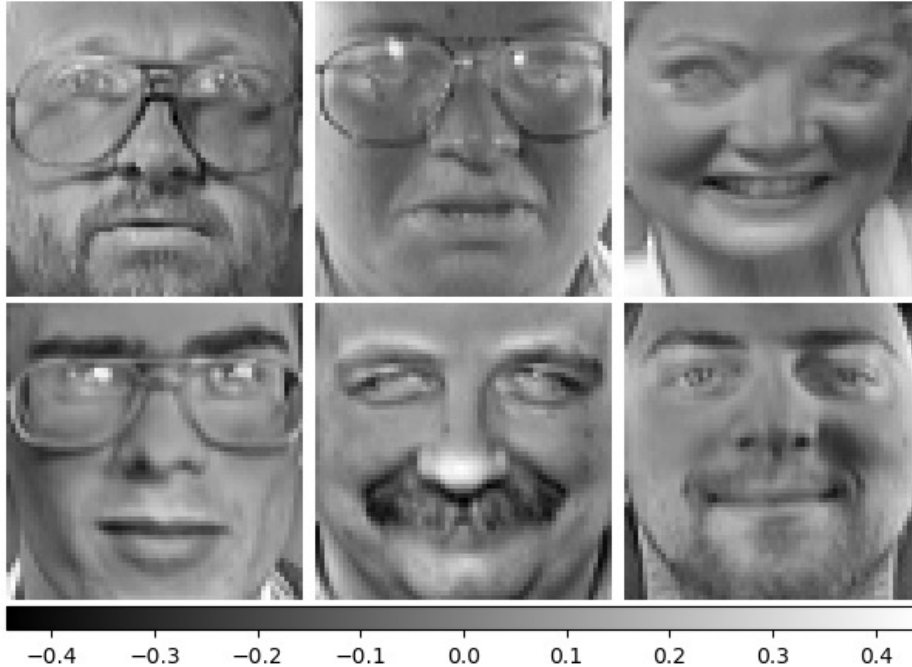
TEMPORAL

- Medical signal processing – fMRI, ECG, EEG
- Brain Computer Interfaces
- Modeling of the hippocampus, place cells
- Modeling of the visual cortex
- Time series analysis
- Financial applications
- Blind deconvolution

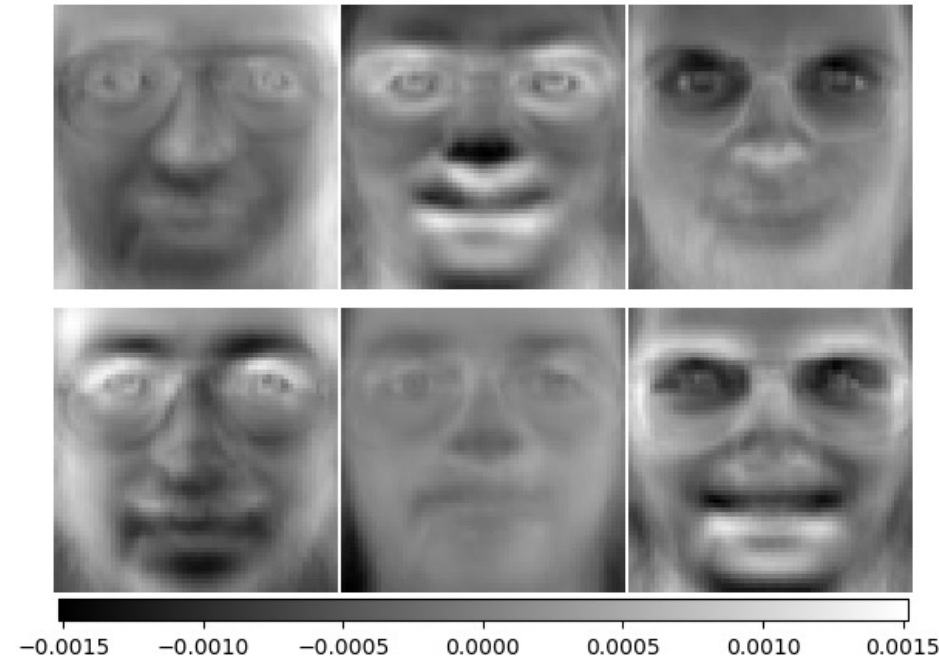


ICA for Image Reconstruction

Faces from dataset



Independent components - FastICA



ICA for Motion Style Components

- Method for analysis and synthesis of human motion from motion captured data
- Provides perceptually meaningful "style" components
- 109 markers, (327dim data)
- Motion capture $\Rightarrow$ data matrix

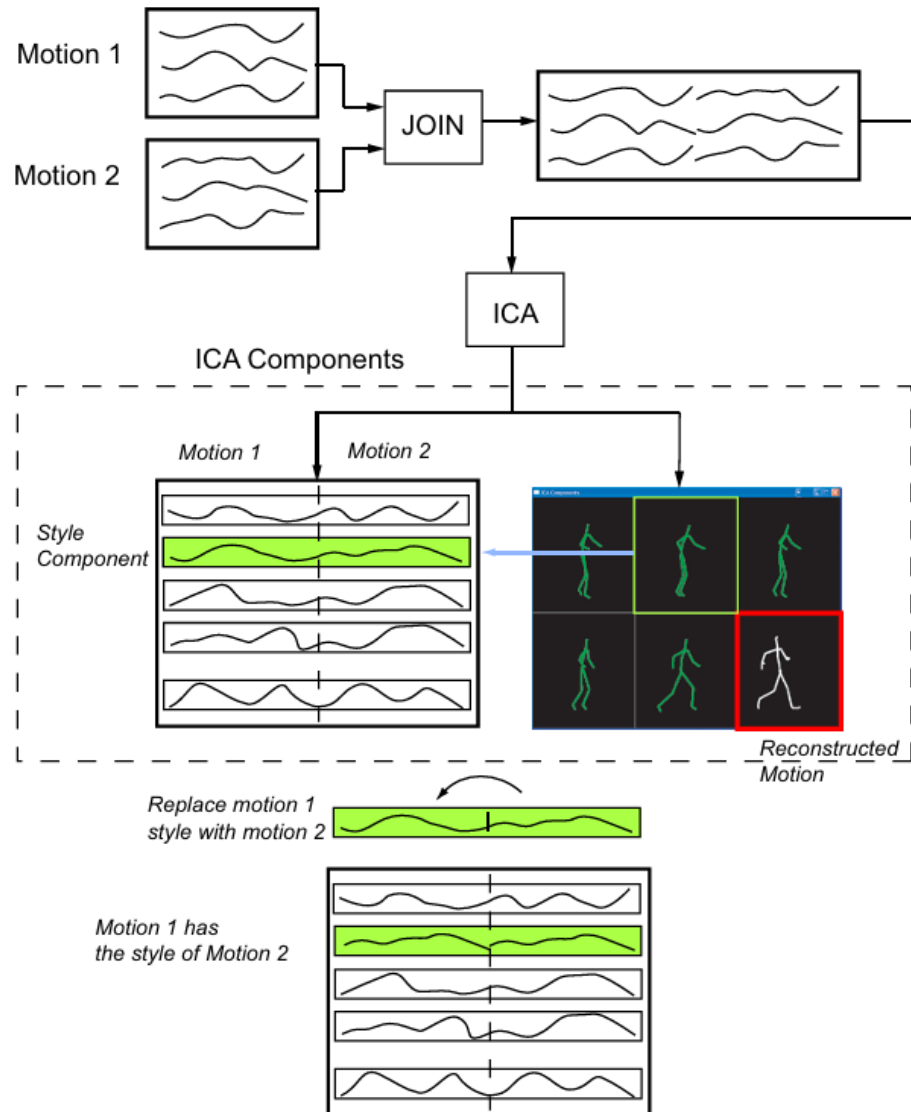
Goal

Find motion style components.

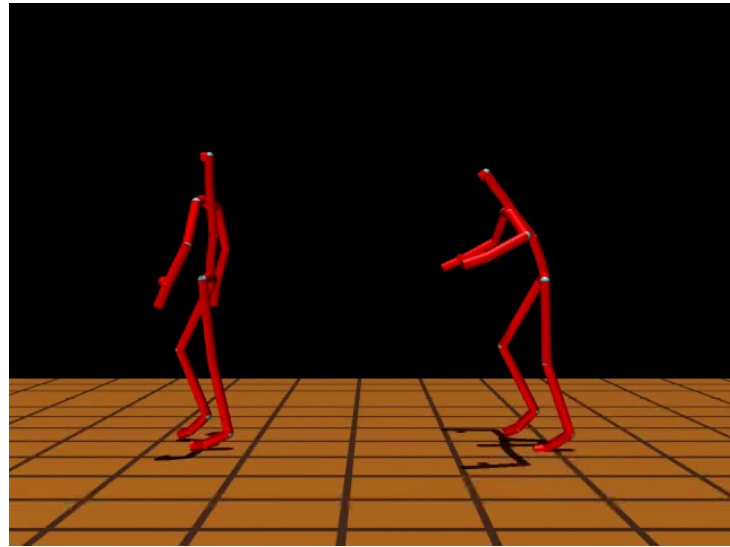
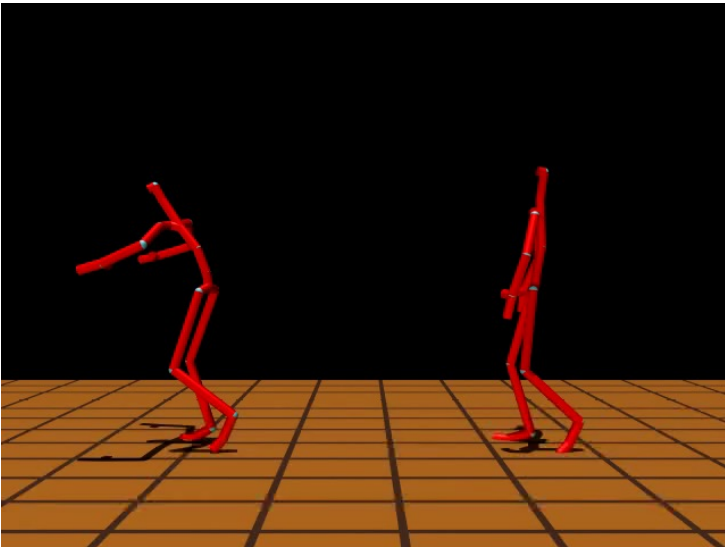
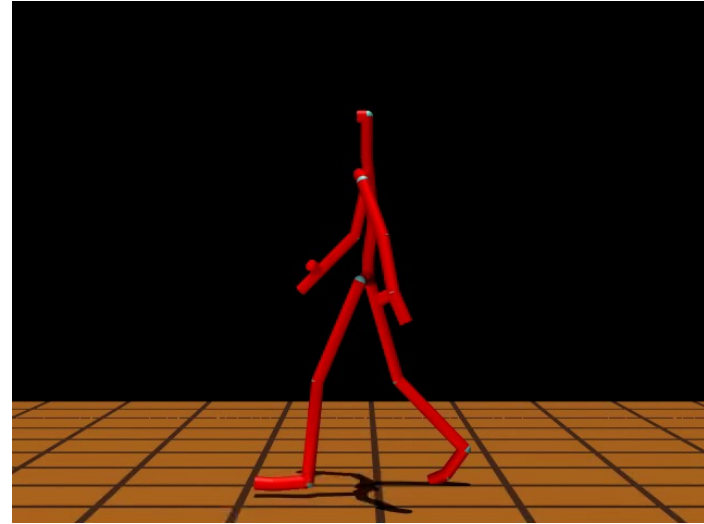
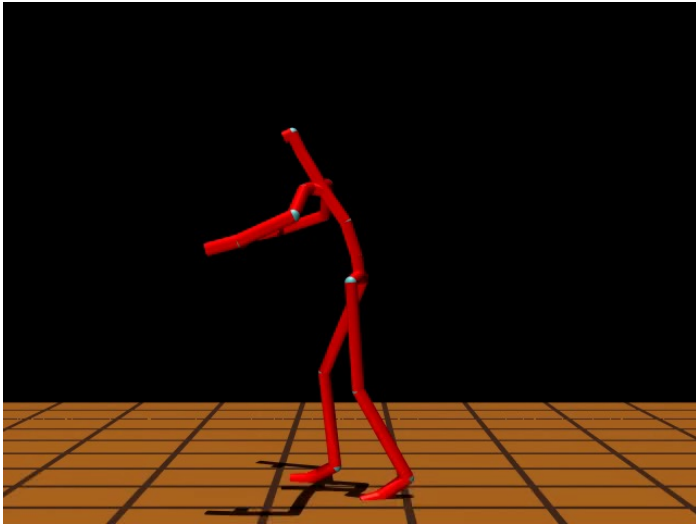
ICA $\Rightarrow$ 6 independent components (emotion, content, ...)



ICA for Motion Style Components



ICA for Motion Style Components



Source: Mori & Hoshino 2002, Shapiro et al 2006, Cao et al 2003



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ICA for Speech Motion Editing

Combined motions

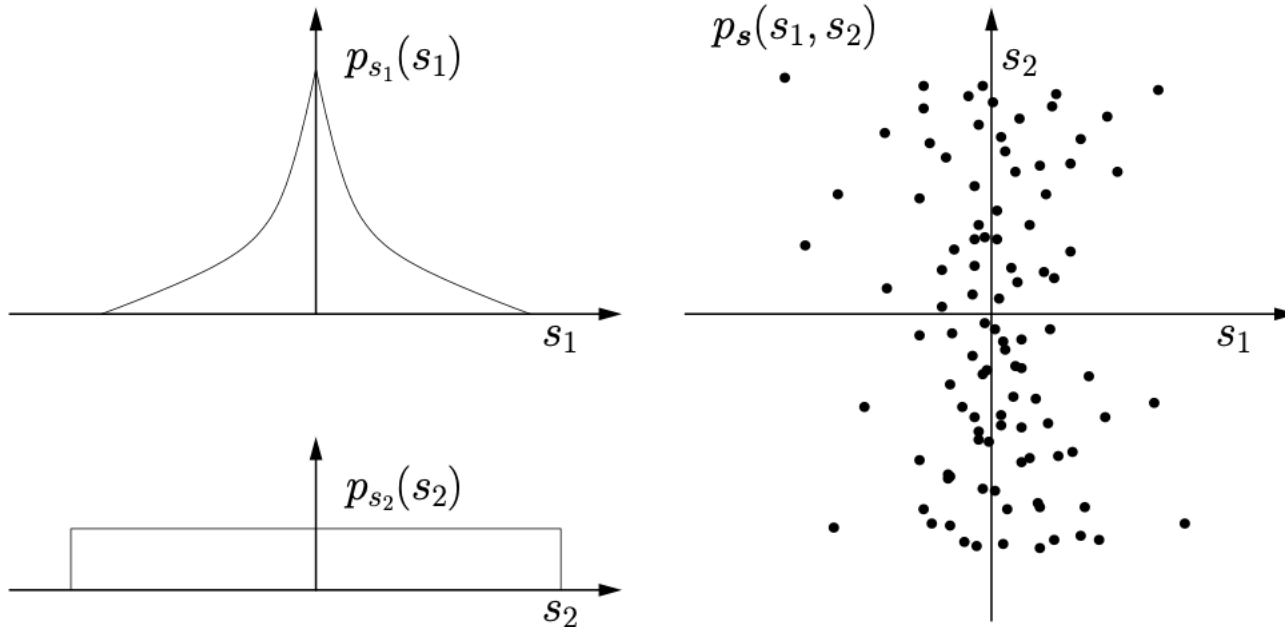
Neutral
Sad
Partial Neutral
Partial Sad



ICA Theory



Statistical (in)dependence



Individual and joint pdfs of two sources

$$S_1, S_2 \text{ are independent} \iff p(s_1, s_2) = p(s_1)p(s_2)$$



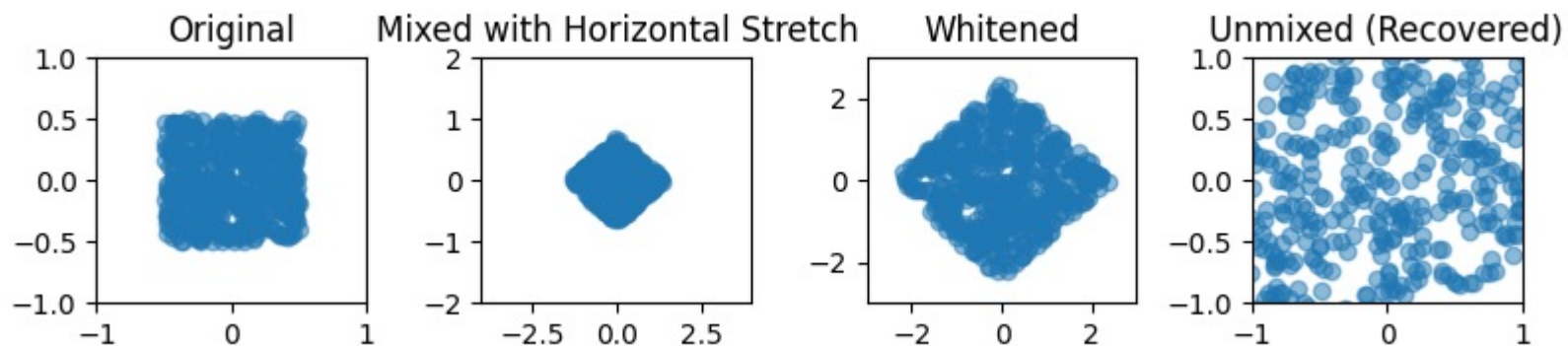
Solving ICA

ICA task: Given x ,

- find y (the estimation of s),
- find W (the estimation of A^{-1})

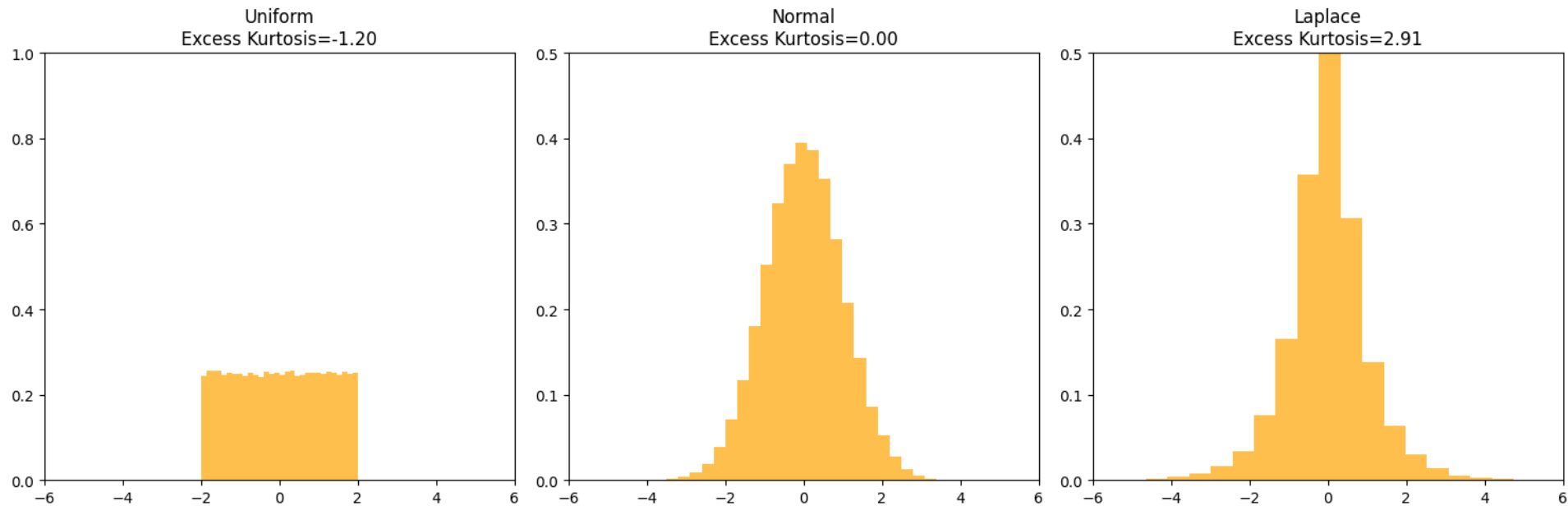
ICA solution: $y = Wx$

- Remove mean, $\mathbb{E}[x] = 0$
- Whitening, $\mathbb{E}[xx^T] = I$
- Find an orthogonal W optimizing an objective function
 - Sequence of 2-d Jacobi (Givens) rotations



Kurtosis

Kurtosis maximization



Central Limit Theorem



CLT 1: [Seeing Theory](#)

CLT 2: [CLT Apps](#)



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Newton–Raphson's method

Goal

Find x^* such that $\phi : \mathbb{R} \rightarrow \mathbb{R}$ and $\phi(x^*) = 0$. What is x^* ?

Linear Approximation (1st order Taylor approx):

$$\phi(x + \Delta x) = \phi(x) + \phi'(x)\Delta x + o(|\Delta x|)$$

Therefore,

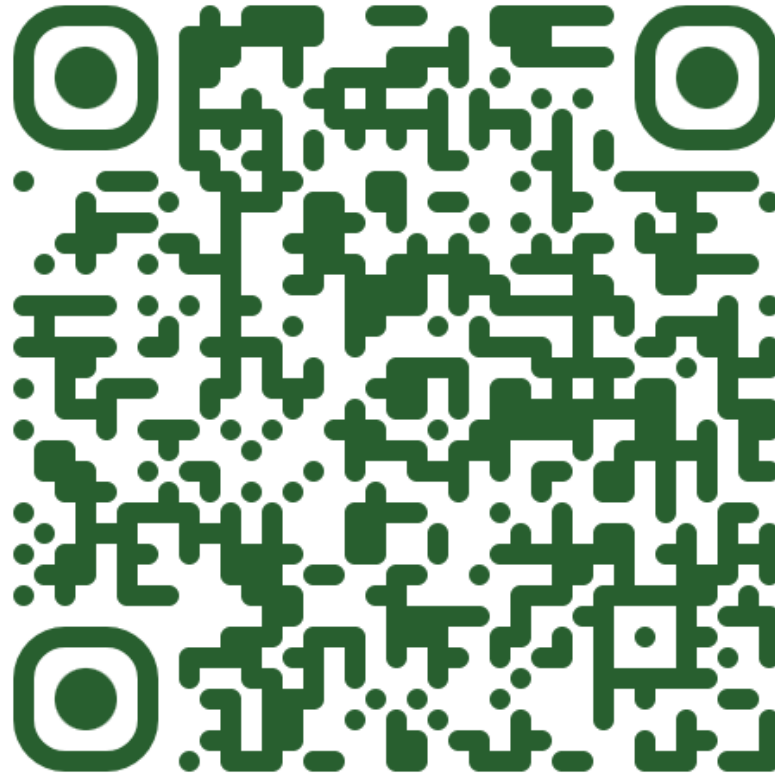
Assuming $0 \approx \phi(x) + \phi'(x)\Delta x$, we get:

$$x^* - x = \Delta x = -\frac{\phi(x)}{\phi'(x)}$$

$$x_{k+1} = x_k - \frac{\phi(x_k)}{\phi'(x_k)}$$



Newton–Raphson's method



Link: [Newton's Method Demo](#)



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The Fast ICA algorithm (Hyvärinen)

Kurtosis maximization

- Given whitened data z
- Estimate the 1st ICA component:

$$y = w^T z, \quad \|w\| = 1, \quad \text{where } w^T = 1^{\text{st}} \text{ row of } W$$

- Maximize kurtosis $f(w) = \kappa_4(y) = \mathbb{E}[y^4] - 3$
- With constraint $h(w) = \|w\|^2 - 1 = 0$

At optimum:

$$\begin{aligned} f'(w) + \lambda h'(w) &= 0^T \\ \Rightarrow 4\mathbb{E}[(w^T z)^3]z + 2\lambda w &= 0 \end{aligned}$$

Solve this equation by Newton-Raphson's method.



Neural Cocktail Party



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Linking Brain, Mind and Behavior

❑ EEG ~ Neural cocktail party

❑ Severe contamination of EEG activity by

- eye movements
- blinks
- muscle
- heart, ECG artifact
- vessel pulse
- electrode noise
- line noise, alternating current (60 Hz)

❑ ICA can improve signal

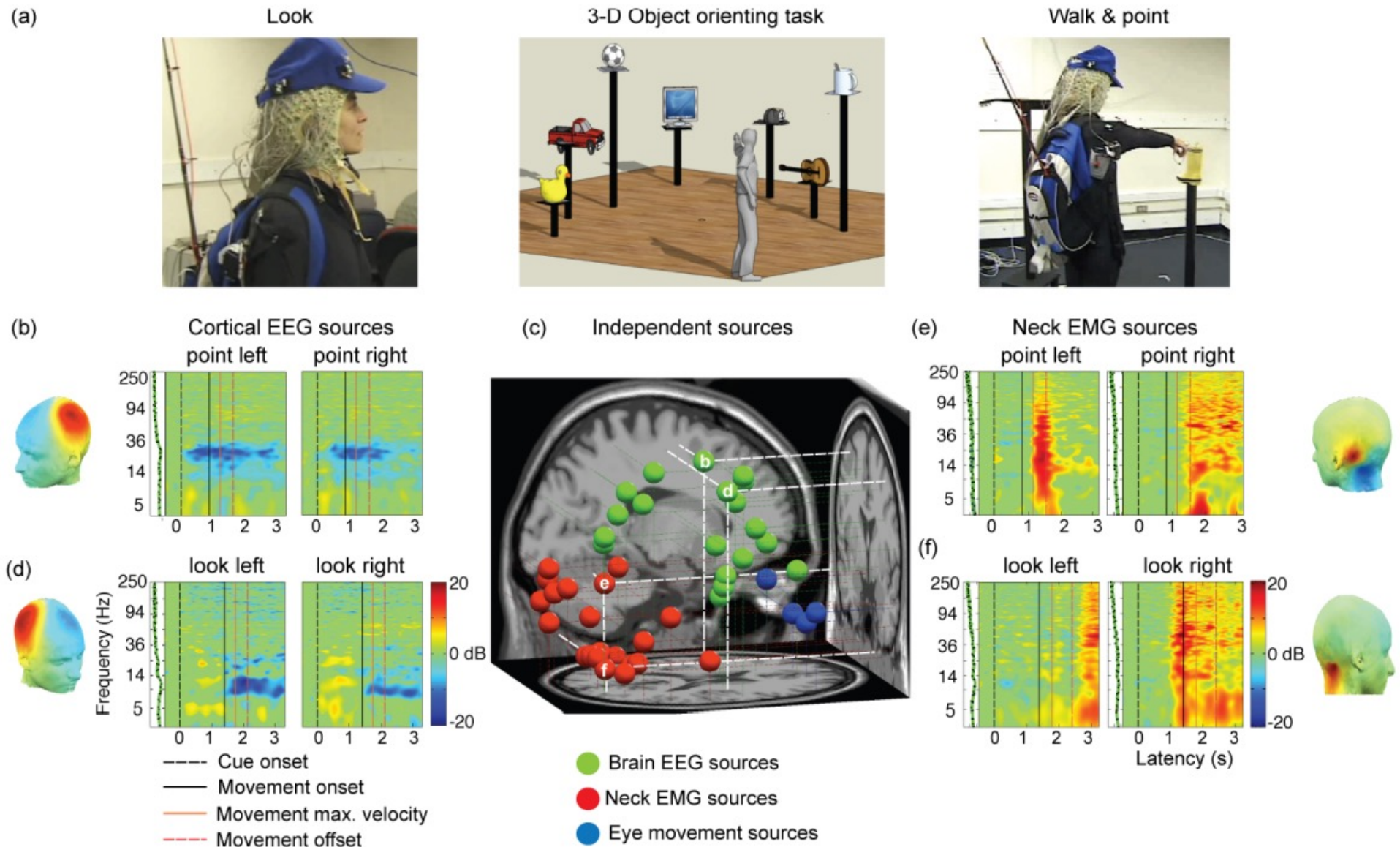
- effectively detect, separate and remove activity in EEG records

from a wide variety of artifactual sources. (Jung, Makeig, Bell, and Sejnowski)

❑ ICA weights (mixing matrix) help find **location** of sources



Linking Brain, Mind and Behavior

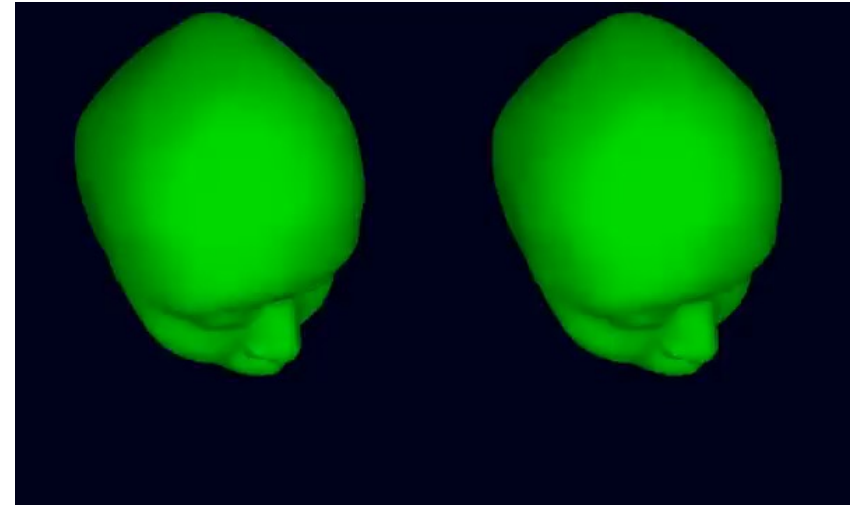
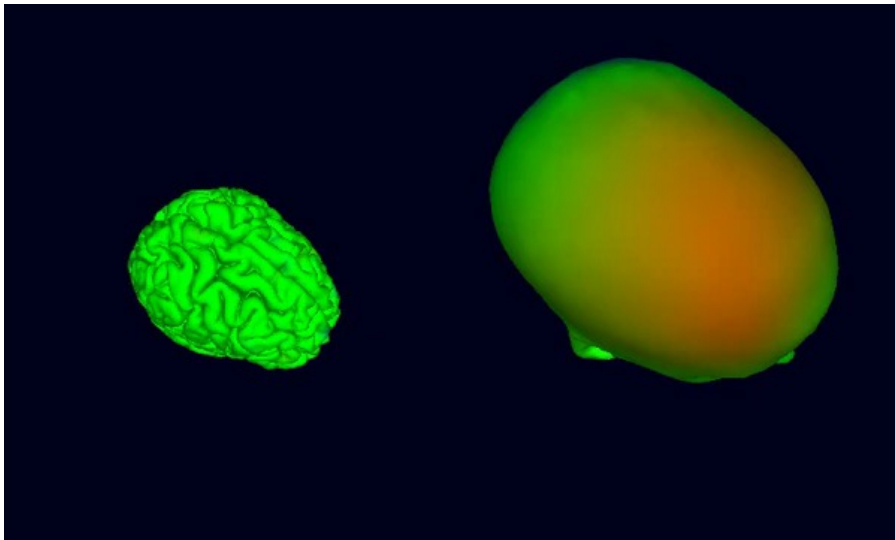
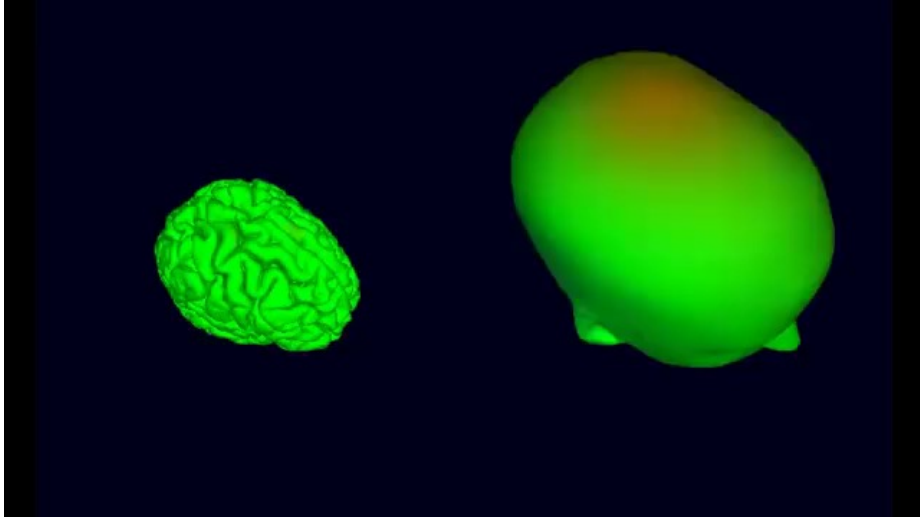


Source: S Makeig, Gramann K, Jung T-P, Sejnowski TJ, Poizner H. Linking brain, mind and behavior, 2009.

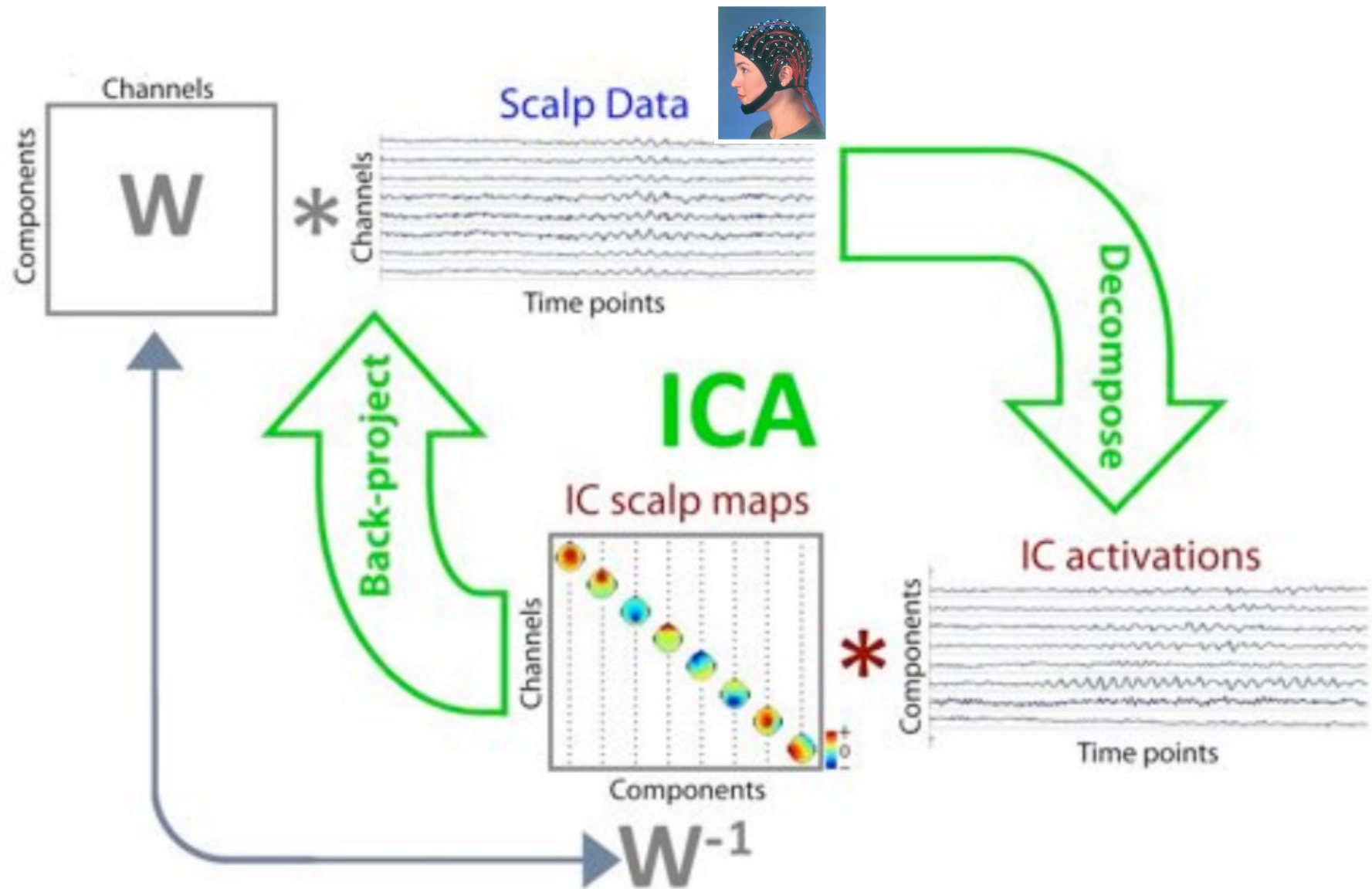


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Linking Brain, Mind and Behavior

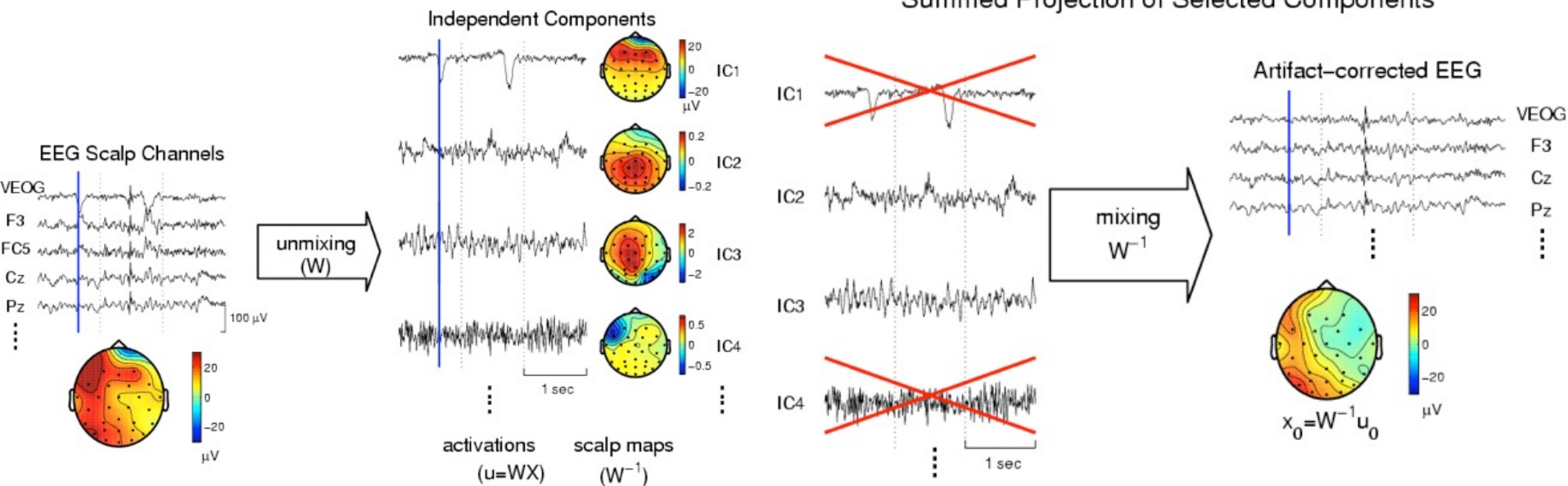


Linking Brain, Mind and Behavior



EEG artifact removal using ICA

ICA decomposition



ICA Label Classification

ICA Demo



Link: [ICA Demo](#)



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